Antenna → waveguide/planer → time domain

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exp-6 - microstrip line

given → $h^{(sub_t)}$ Calculate w from VBA macros → Calculate → analytical impedance

then $\frac{w}{h}$ Calculate

$\frac{w}{h} \le 1$ $\quad Z_0 = \frac{60 \ln\left[\frac{8h}{w} + \frac{w}{4h}\right]}{\sqrt{\varepsilon_{eff}}}$

$\frac{w}{h} > 1$ $\quad Z_0 = \frac{120\pi}{\sqrt{\varepsilon_{eff}}\left[\frac{w}{h} + 1.393 + 0.667 \ln\left(\frac{w}{h} + 1.444\right)\right]}$

sub_t = 0.5mm, 0.479 = w,

Model

→ पहले substrate बनायेंगे

→ ~~select~~ model → choose brick → name it substrate → -board/2 board/2, -board/2 board/2, -sub_t 0

→ then select material

board-size भी given होगा

→ अब ground बनायेंगे

flip करो (पीछे की side मांगो)

Pick face → Extrude → name it ground → material (PEC)

height → let say (pec_t) → set 0.01mm (कम रखनी है)

→ now back to front

go to curves → select line → draw line → 0 -board/2, 0 board/2

→ Trace from curve, double click on line & then Enter & set dimensions → name it microstrip

Thickness → same as ground (pec_t)

width = w (जो अभी calculate करी थी)

material → same as ground (PEC)

→ now define ports at the edges of the line, ~~go~~ ~~to~~ pick face

& go to VBA macros, Calculate port extension coefficient

& Construct port

do it for both edges

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→ simulation → boundaries set करनी है।

x_{min}, x_{max}, z_{min} → electric ($E_t = 0$)
बाकी open (add space)

→ only this experiment we have to do in freq. domain
→ setup solver को freq. domain पर करके
normalize s parameter check box को tick करके run करदे।

Results (Axis marker लगा के value exact पता लग जाएगी किस freq. पर)

→ s parameter → same (S11, S22) → जितना -ve उतना बढ़िया
S21, S12 → जितना 0 के close उतना बढ़िया

→ VSWR → जितना 1 के close उतना बढ़िया

→ Field monitor add करदो efield, h field जिस freq. पर design कर रहे हो वो select करके वापस run करदो

→ Reference impedance उतना ही आना चाहिये जितने का बना रहे हैं
ex 50 Ω का है तो 50

→ effective dielectric constant ≠ material (Alumina-9.4) because we used PEC also, so overall ε_{eff} will be less than ε_r

→ efield, h field → दोनों center से ही जाने चाहिये।

Trick

s parameter smith chart में center में dot आना चाहिये।

exp-7 matched & unmatched load

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→ delete any one port

→ अब load लगाना है।

फिर pick edge (2 edges) → lumped element लगा

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

for matched load $\Gamma = 0$ $(Z_L = Z_0)$ so $VSWR = 1$

Results

Sparameter → S11 → कितना -ve रहना चाहिए

VSWR → 1 के close

arrow की जगह contour select for best view of e-field & H-field

→ Unmatched

→ lumped element → edit properties (R, L, C को set करे delete)

$$X_L = 2\pi f L \quad , \quad X_C = \frac{1}{2\pi f C}$$

$$Z_L = R + j(X_L - X_C)$$

if $Z_L >> Z_0$ $\Gamma \approx 1$ & $VSWR = \infty$

Very high आना चाहिये।

Sparameter smith chart में 2.4 GHz पर

Capacitive Z_L के लिए upper side आना चाहिये

Inductive Z_L —— // lower —//——

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exp-8

Patch Antenna

→ पहले substrate बनायेंगे

brick select करो → -sl/2 sl/2
→ -sw/2 sw/2
→ -sh 0

select material → Aluminum (9.0)

sh → substrate thickness

website → microstrip patch Antenna Calculator (pasternack)

give → Er, dielectric height (sub_thickness), operation freq.

it will give width & length of patch [Remember not of substrate]

pw = 27.39 20.35 = pl

→ for substrate sl & sw take them greater than 6 from width & length of patch.

→ अब ground बनायेंगे

→ flip it 2

transform WCS → sw को -sh करके OK

→ now pick brick, name it ground

→ -sl/2 sl/2
→ -sw/2 sw/2
→ -mt 0

material → copper (annealed)

mt = 0.035 mm (0.01 से 0.05 के बीच ले तो कुछ भी)

→ अब patch बनाना है

~~flip again~~

WCS को ऊपर लाना है तो substrate wala face pick करके align WCS से WCS ऊपर आ जायेगा।

mm to mil, 39.37 से multiply

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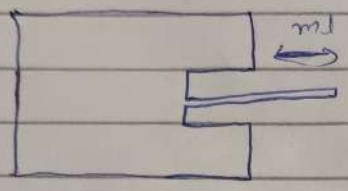
select brick → name it patch

$-pl/2$	$pl/2$
$-pw/2$	$pw/2$
0	mt

pl & pw values are calculated through website

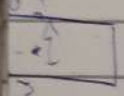


Transform
DU → $-pl/2$

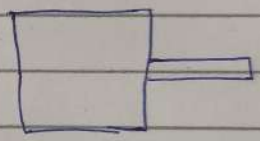
brick add → name it microstrip

0	ml
$-mw/2$	$mw/2$
0	mt

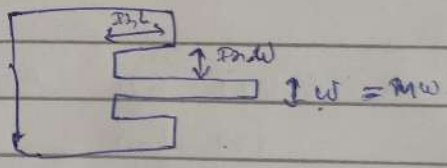
$ml = sl/2 - pl/2$,
mw → Calculate from VBA macros



→ ~~Select brick~~



ab cut karna है,



Inl calculate from website
Atlium,
freq. = 300

Inw = (1-2) w should be,

→ port Construct करो
→ then run (check normalize S parameters)

→ Results, add field monitors E field, H-field, farfield

verify → E field & H field will be at edges

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Exp-1 Rectangular waveguide

$-a/2 \quad a/2$

$-b/2 \quad b/2$

$0 \quad l$

eplane tee → $-a/2 \quad a/2$

$b/2 \quad 3b/2$

$l/2 - \frac{b}{2} \quad l/2 + b/2$

H plane tee →

---

patch antenna array

→ select all ground, substrate, patch, microstrip
Transform → Translate → copy & paste
boolean add → substrate, ground plane
ports define करके run

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$g = 0.005\lambda$, to $0.02\lambda_0$

final Horn Antenna

↳ Make waveguide

↳ Bring net on tz point

↳ $\lambda = \frac{c}{f} \Rightarrow c = \frac{c}{\sqrt{\varepsilon_r}}$

↳ now shift net from $1.5\lambda_0 - 2\lambda_0$

$\Rightarrow$ take 60mm

↳ $\text{gain} = \frac{4\pi \text{Area}}{\lambda^2}$ $\Rightarrow$ gain se area निकालो

default 15dB

Lit in linear

$(10^{\frac{dB}{10}})$

$\frac{A}{B} = \frac{a}{b} \times$ waveguide dim. → ①

horn dimens.

$A \times B$ = area → ②

↳ find A & B

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Make brick of A & B (Put z = 0)

↳ Pick faces to join → Model → loft
↓
ok

↳ in comp, select all 3 solids, Boolean add

↳ Pick face wg का बाहर Nd horn का बाहर → Model → shape tools → shell solid or thicken sheet
↳ thickness = 1
⇒ ok

↳ Pick face ⇒ wg port
↳ define freq.
↳ run in freq domain